

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

# Chapter 8: Introduction to Trigonometry

Previous year board questions, topic-wise: 27 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Trigonometric ratios

1. If  $\sin \theta = \frac{a}{b}$ , write  $\cos \theta$  in terms of  $a$  and  $b$ . (a)  $\frac{b}{\sqrt{b^2 - a^2}}$  (b)  $\frac{b}{a}$  (c)  $\frac{\sqrt{b^2 - a^2}}{b}$  (d)  $\frac{a}{\sqrt{b^2 - a^2}}$   
CBSE 2026, 1 mark
2. If  $\tan \theta = \frac{24}{7}$ , find  $\sin \theta + \cos \theta$ . CBSE 2026, 2 marks
3. If  $\cot \theta = \frac{7}{8}$ , evaluate  $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$ . CBSE 2026, 2 marks
4. If  $2 \tan A = 3$ , find the value of  $\frac{4 \sin A + 3 \cos A}{4 \sin A - 3 \cos A}$ . (a)  $\frac{7}{\sqrt{13}}$  (b)  $\frac{1}{\sqrt{13}}$  (c) 3 (d) does not exist  
CBSE 2023, 1 mark
5. If  $\sin \alpha = \frac{1}{\sqrt{2}}$  and  $\cot \beta = \sqrt{3}$ , find  $\operatorname{cosec} \alpha + \operatorname{cosec} \beta$ . CBSE 2023, 2 marks
6. If  $4 \tan \theta = 3$ , evaluate  $\frac{4 \sin \theta - \cos \theta + 1}{4 \sin \theta + \cos \theta - 1}$ . CBSE 2018, 3 marks

## 2 Ratios of standard angles

7. If  $\cos A = \frac{1}{2}$ , find the value of  $\sin^2 A + 2 \cos^2 A$ . (a)  $\frac{3}{2}$  (b)  $\frac{5}{4}$  (c)  $-1$  (d)  $\frac{1}{2}$  CBSE 2026, 1 mark
8. Find the acute angle  $\theta$  for which  $7 + 4 \sin \theta = 9$ . CBSE 2025, 1 mark
9. Given that  $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$ , find the value of  $x + 2y$ . CBSE 2025, 2 marks
10. Evaluate  $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$ . CBSE 2025, 2 marks
11. Given  $\cos(\alpha + \beta) = 0$ , find  $\cos\left(\frac{\alpha + \beta}{2}\right)$ . (a)  $\frac{1}{\sqrt{2}}$  (b)  $\frac{1}{2}$  (c) 0 (d)  $\sqrt{2}$  CBSE 2024, 1 mark
12. Find the value of  $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$ . CBSE 2024, 2 marks
13. For  $A = 60^\circ$  and  $B = 30^\circ$ , verify that  $\sin(A + B) = \sin A \cos B + \cos A \sin B$ . CBSE 2024, 2 marks
14. Evaluate  $\frac{3}{4} \tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ$ . (a)  $-1$  (b)  $\frac{5}{6}$  (c)  $-\frac{3}{2}$  (d)  $\frac{1}{6}$  CBSE 2023, 1 mark

### 3 Trigonometric identities

15. Given  $x = h + a \cos \theta$  and  $y = k + b \sin \theta$ , prove that  $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$ .  
CBSE 2026, 3 marks
16. Prove that  $\frac{\tan A}{1 + \sec A} - \frac{\tan A}{1 - \sec A} = 2 \operatorname{cosec} A$ . CBSE 2026, 3 marks
17. Simplify  $\tan^2 \theta - \left(\frac{1}{\cos \theta} \times \sec \theta\right)$ . CBSE 2025, 1 mark
18. Prove that  $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$ . CBSE 2025, 3 marks
19. Prove that  $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2 \sin^2 A - 1}$ . CBSE 2025, 3 marks
20. Given  $\sec \theta - \tan \theta = m$ , find  $\sec \theta + \tan \theta$ . (a)  $1 - \frac{1}{m}$  (b)  $m^2 - 1$  (c)  $\frac{1}{m}$  (d)  $-m$  CBSE 2024, 1 mark
21. Prove the identity  $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$ . CBSE 2024, 3 marks
22. If  $\sin \theta + \cos \theta = \sqrt{3}$ , find the value of  $\sin \theta \cos \theta$ . CBSE 2023, 2 marks
23. Prove that  $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$ . CBSE 2023, 3 marks
24. Prove that  $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$ . CBSE 2019, 3 marks
25. Prove that  $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$ . CBSE 2019, 3 marks
26. Prove that  $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{1}{\sec A - \tan A}$ . CBSE 2019, 4 marks
27. Prove that  $\frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \tan A$ . CBSE 2018, 4 marks

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